

AutoML and such

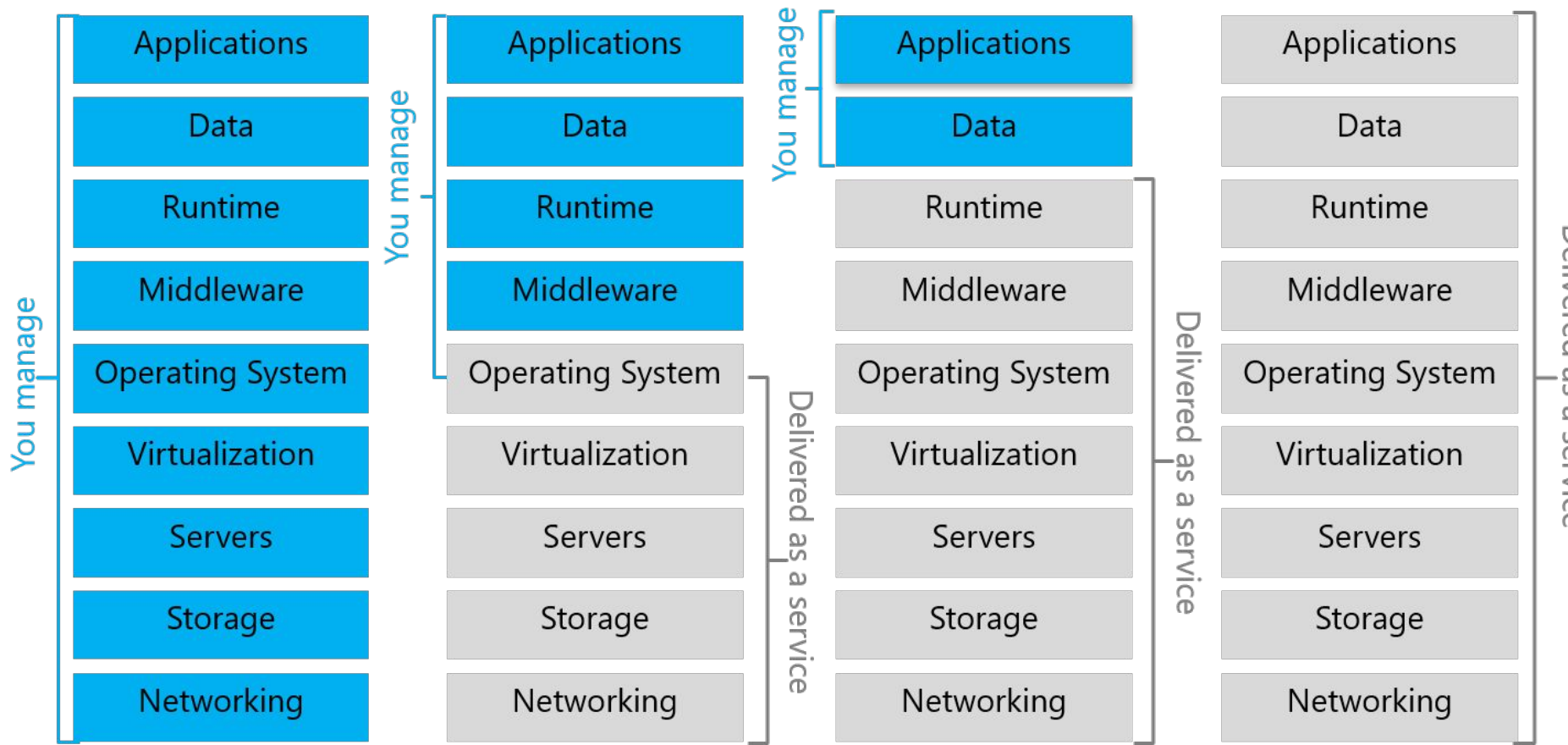
But before that..

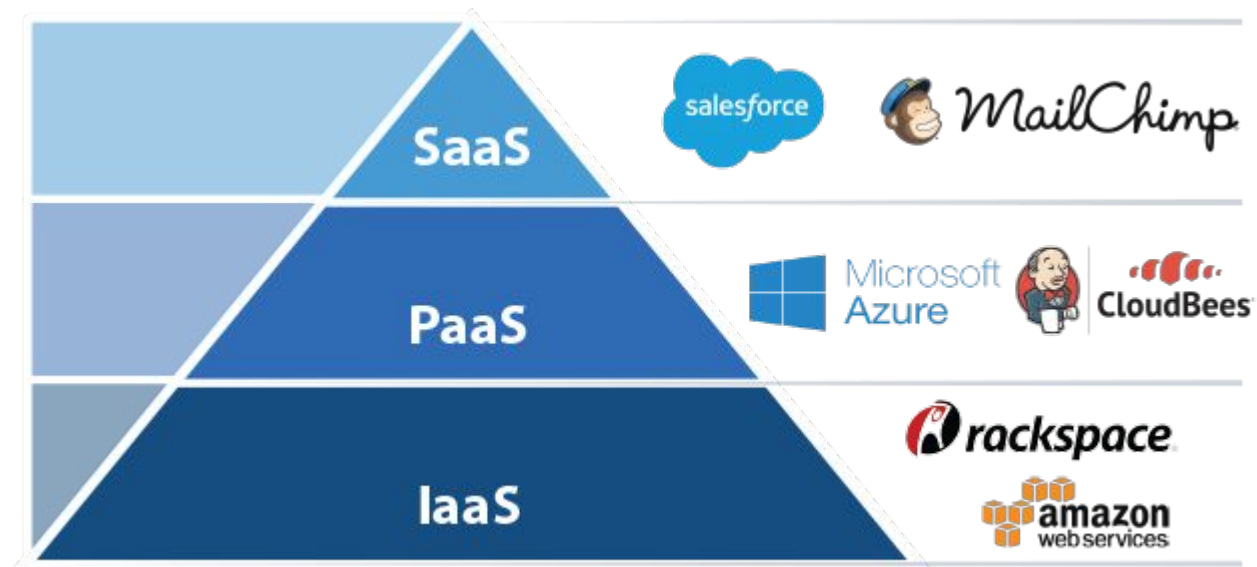
Traditional IT

Infrastructure (as a Service)

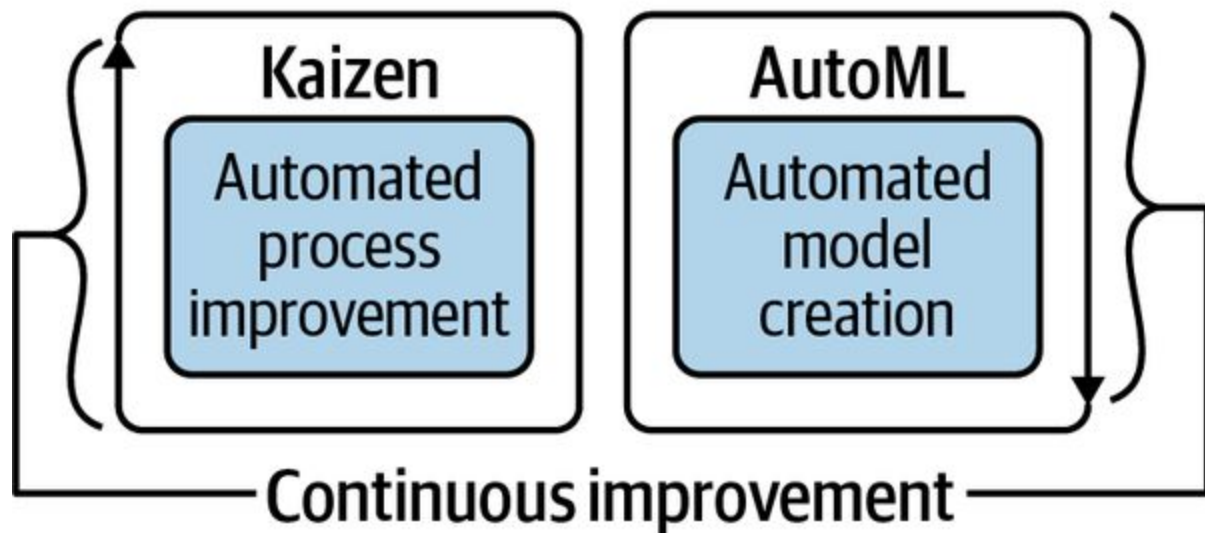
Platform (as a Service)

Software (as a Service)





KaizenML or AutoML?



Feature Stores

- i.e., a registry of high-quality machine learning inputs and the ability to monitor data for drift and register and serve out ML models
- Allows users to add features they built into a shared Feature Store.
- Once features are in the Feature Store, they are easy to use in training and prediction.
- Uber Engineering:
<https://www.uber.com/en-PK/blog/michelangelo-machine-learning-platform/>
-

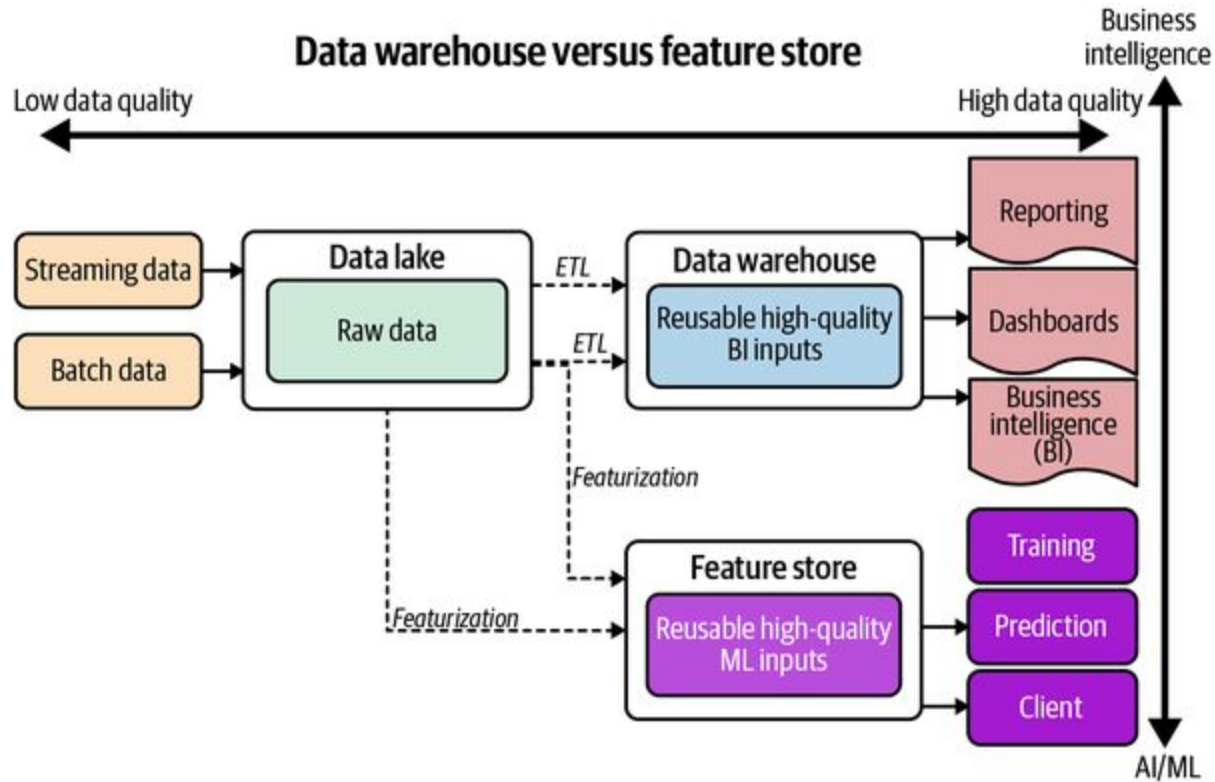
The Problem - Data Silos and Inconsistency

Before we define what a Feature Store is, let's understand the problem it solves. In many organizations, data science teams work in silos. Team A calculates customer lifetime value one way; Team B calculates it slightly differently. When it's time to deploy a model, the data used for live predictions might not be calculated in the exact same way as the data used for training. This leads to a critical problem called **train-serve skew**, where your model's performance degrades in production because the live data doesn't match the training data. It's a recipe for silent failure.

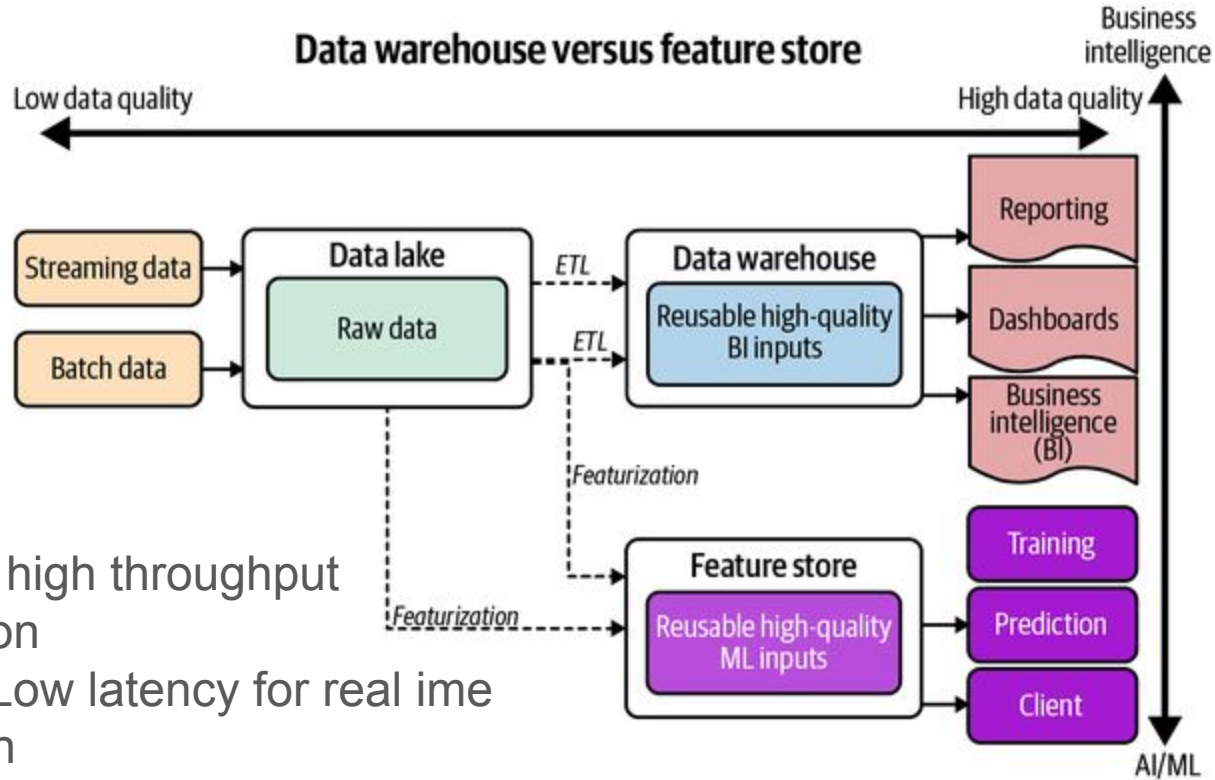
Data Warehouses vs Feature stores

A data warehouse feeds into business intelligence systems at a high level, and a Feature Store provides inputs into an ML system

Data warehouse versus feature store

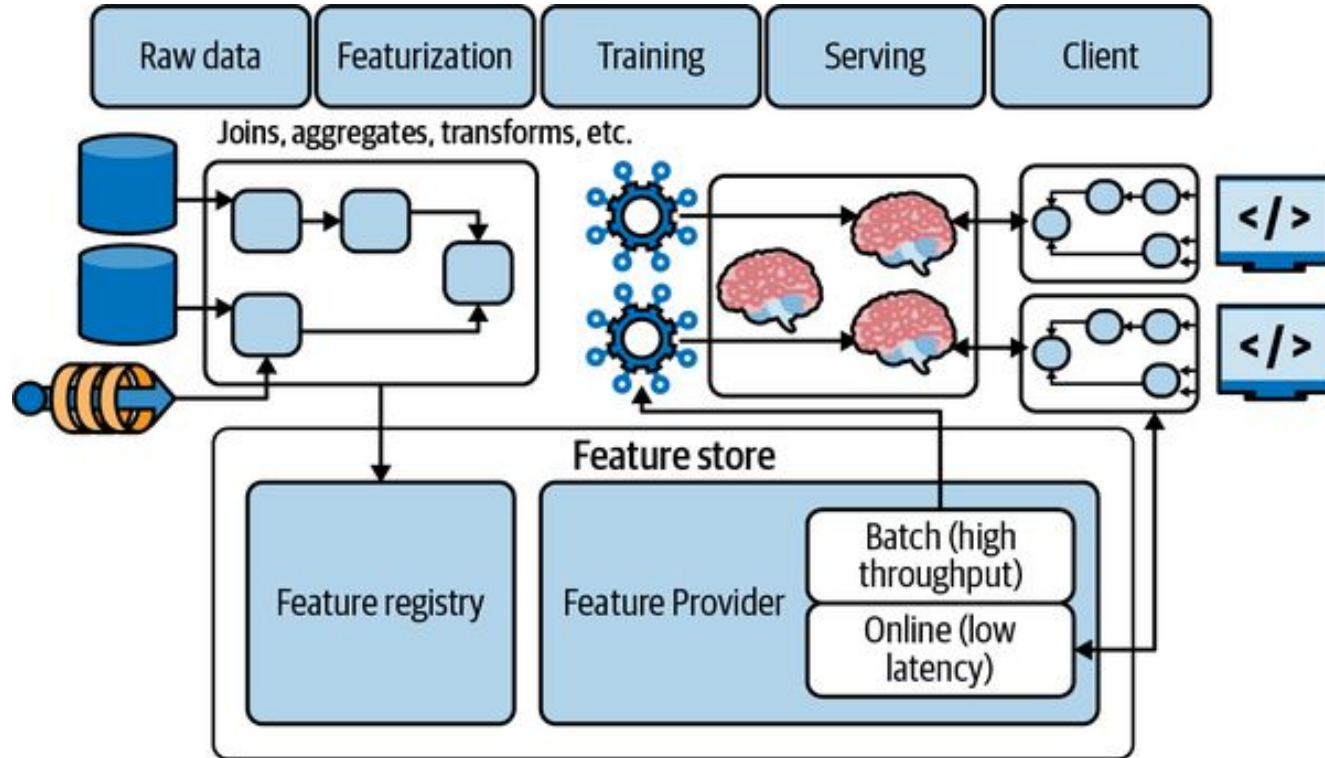


Data warehouse versus feature store



1. Training: high throughput connection
2. Testing: Low latency for real time prediction

Example: Databricks



Helps in Discovery

Monitoring Data Drift

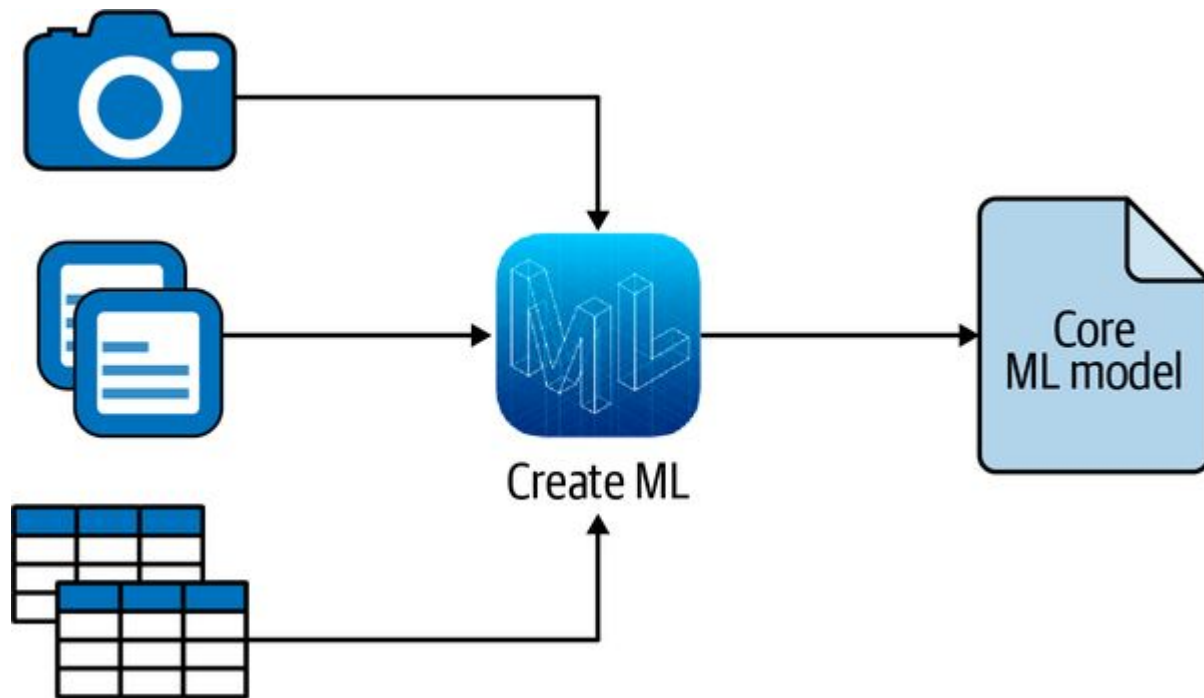
Solves the train-serve skew problem

AutoMIs in Action..

Automates basically everything...

- Data Preparation 
- Feature Engineering 
- Model Selection 
- Hyperparameter Optimization 

Apple



All

New

Image

Video

Motion

Sound

Text

Table

Choose a Template



Image Classification



Object Detection



Style Transfer



Action Classification



Activity Classification



Sound Classification



Text Classification



Word Tagging



Tabular Classification



Tabular Regression



Recommendation

Image Classification

Classify the dominant object or scene in an image.

Cancel

Previous

Next

Google

Google Cloud Platform cloudai Search products and resources

Vision flowers LABEL STATS EXPORT DATA

Dashboard Datasets Models

IMPORT IMAGES TRAIN EVALUATE TEST & USE Multi-Label Classification

All images 3,667
Labeled 3,667
Unlabeled 0

1 selected ASSIGN LABELS DELETE

Select all

Filter Filter labels +

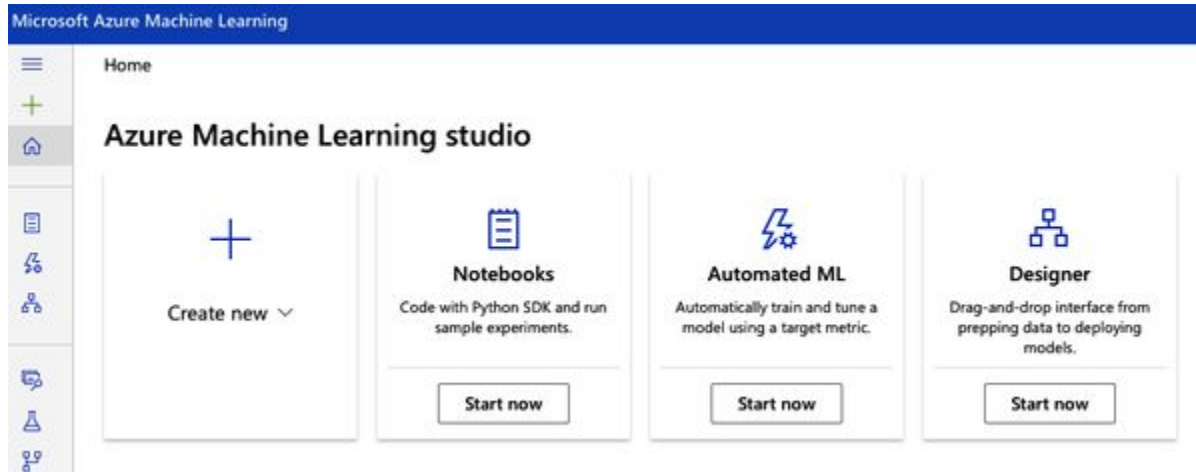
daisy	633
dandelion	898
roses	641
sunflowers	697
tulips	799

ADD NEW LABEL

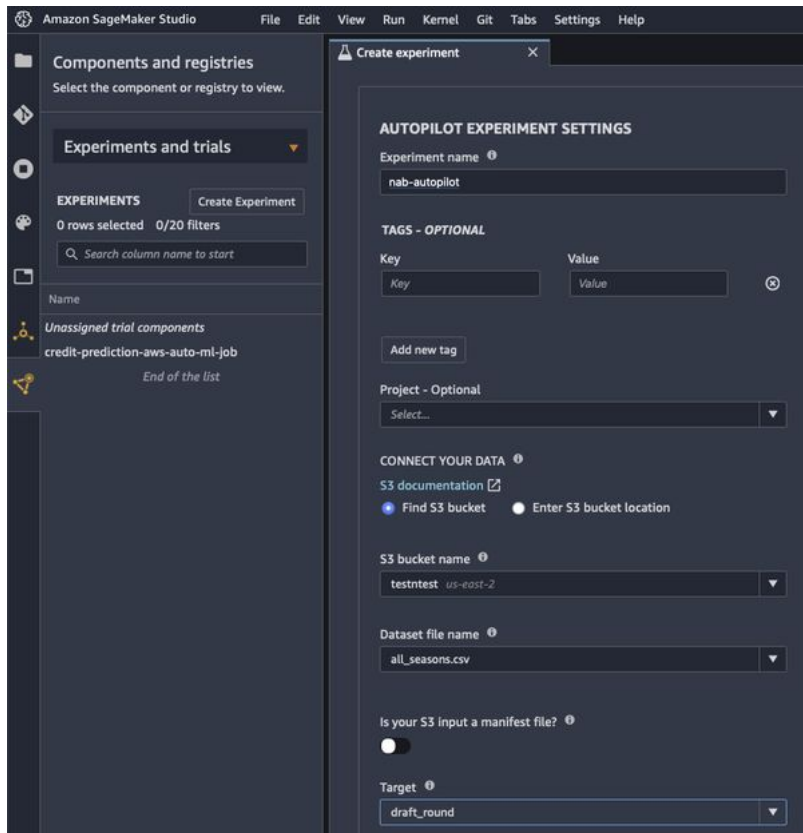
daisy(1) daisy(1) daisy(1)

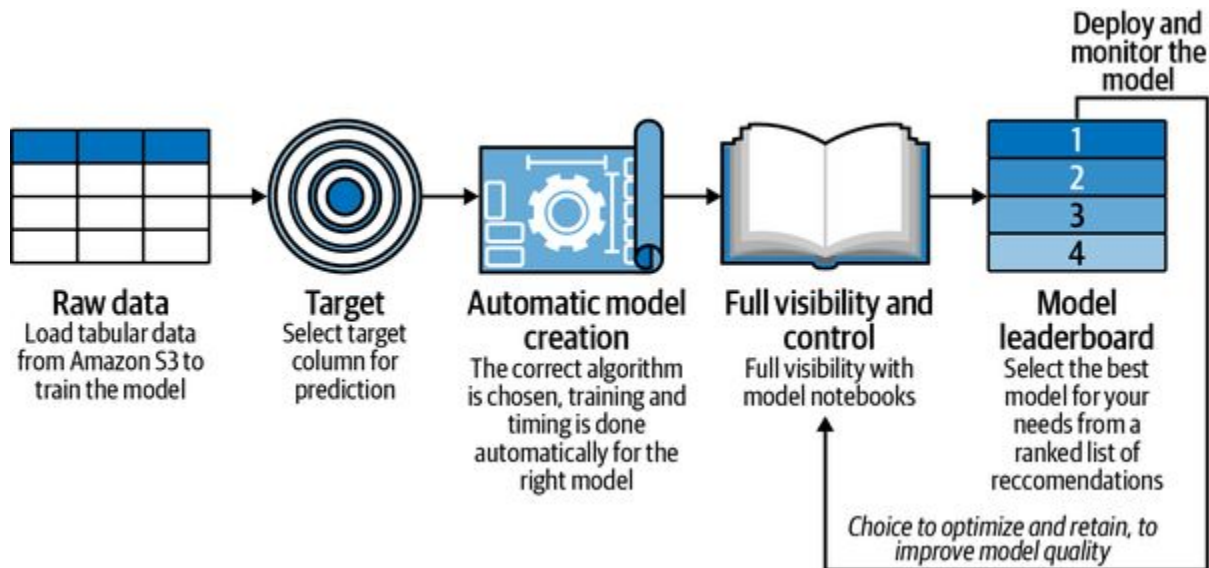
daisy(1) daisy(1) daisy(1)

Azure



AWS: Sagemaker





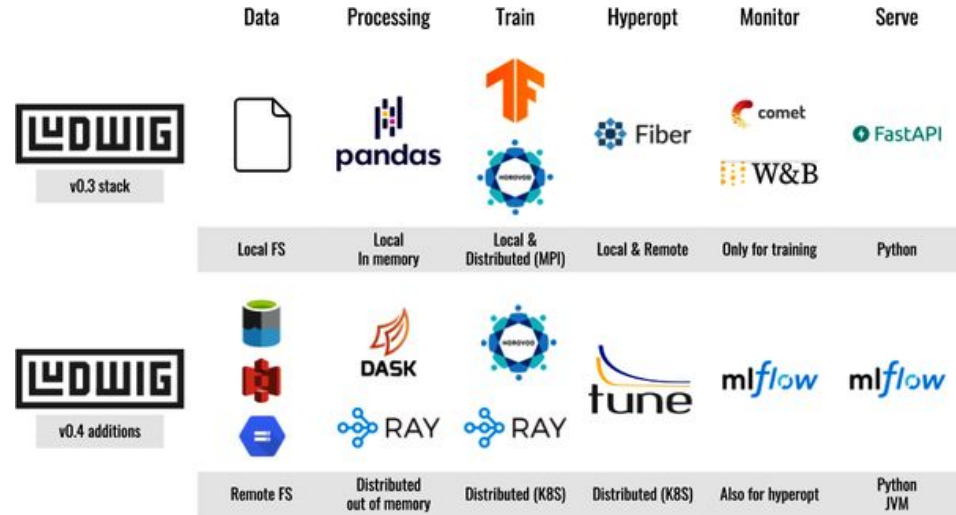
Open Source options do exist

- Ludwig
- FLAML

FLAML - Fast and Lightweight AutoML



Figure 5-40. FLAML from Microsoft Research



Spoiler Alert: The world is your oyster.. Choose your weapons of choice